Math 445/545. Winter 2026. Practice problems for the Midterm 2.

The problems below illustrate various techniques you are supposed to know for the final. Some of the problems below are more difficult than the ones you will see on the midterm.

Problem 1. In the ring $\mathbb{Z}$, find a positive integer a such that $\langle a \rangle = \langle 6 \rangle + \langle 8 \rangle$ (here $+$ stands for the sum of ideals).

Problem 2. Is the ideal $N = \langle 35 \rangle \subset \mathbb{Z}$ maximal? How many ideals are strictly in between of N and $\mathbb{Z}$?

Problem 3. Let $N = \langle 2 \rangle \subset \mathbb{Z}$. Prove that $N[x] \subset \mathbb{Z}[x]$ is an ideal of $\mathbb{Z}[x]$. Is the ideal $N[x]$ maximal? Is it prime?

Problem 4. How many elements are in $\mathbb{Z}[i]/\langle 3+i \rangle$?

Problem 5. Prove that $\langle 2+2i \rangle$ is not a prime ideal of $\mathbb{Z}[i]$. What is characteristic of $\mathbb{Z}[i]/\langle 2+2i \rangle$?

Problem 6. Prove that $\mathbb{Z}_2[x]/\langle x^2+x+1 \rangle$ is a field. How many elements are in this field?

Problem 7. Assume R is PID and let $N \subset R$ be an ideal. Prove that any ideal of R/N is principal. True/False: under the assumptions above, R/N is always PID.

Problem 8. Let $\phi : \mathbb{Z}[i] \to \mathbb{Z}_{13}$ be a unital ring homomorphism (such homomorphisms do exist). What are possible values of $\phi(i)$?

Problem 9. Show that the rings $\mathbb{Z}[x]/(x^2+2x+3)$ and $\mathbb{Z}[\sqrt{-2}]$ are isomorphic.

Problem 10. Show that the rings $\mathbb{Z}[\sqrt{-2}]$ and $\mathbb{Z}[i]$ are not isomorphic.